

PyTorch Project to Build a GAN Model on MNIST Dataset

Overview

Have you ever wondered how photo editing applications are able to convert night images to day images? Also, you must have come across some photos of people who do not exist in the world. One of the powerful tools to achieve this is Generative Adversarial Networks also known as GAN. The GANs are mainly used in image-to-image translation and to generate photorealistic images which even a human fails to identify as fake or true.

In this project, we start by introducing Generative Adversarial Networks. The PyTorch framework is used to build the GAN model on the MNIST dataset. Finally, we learn how to use the Generator model for generating new images of digits.

Aim

1. To understand the theory behind Generative Adversarial Network
2. To build Generative Adversarial Networks in PyTorch

Tech Stack

- Language: Python
- Libraries: torch, torchvision, numpy, matplotlib

Approach

1. Introduction to Generative Adversarial Network
2. Introduction to Generator and Discriminator
3. Loss function in GAN
4. Building Model on PyTorch
5. Model training on Google Colab
6. Generating fake images with Generator

Prerequisites

1. [Learn to Build Generative Models Using PyTorch Autoencoders](#)
2. [Learn How to Build PyTorch Neural Networks from Scratch](#)

Modular code overview

```
data
  |_data_utils.py

model
  |_gan.py

run.py

train.py

GAN_Notebook_colab.ipynb

requirements.txt
```

Once you unzip the modular_code.zip file, you can find the following folders within it.

1. data
 2. model
 3. run.py
 4. train.py
 5. requirements.txt
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1. The data folder contains data_utils.py file which is used to download and transform the data. It will download and store the data in the respective folder
 2. The model folder contains gan.py
 3. The train.py files contain the code for model training. The run.py is the main file in which all functions are called
 4. GAN_Notebook_colab.ipynb is used to train the model on colab
 5. The requirements.txt file has all the required libraries with respective versions. Kindly install the file by using the command **pip install -r requirements.txt**
Note: please use cuda versions of torch if CUDA is available

Project Takeaways

1. What is Generative Adversarial Network?
2. Use of Generative Adversarial Network
3. What is Generator?
4. What is Discriminator?
5. Understanding architecture of Generative Adversarial Network
6. The loss function of Generator
7. The Loss function of Discriminator
8. How to download and transform data in PyTorch?
9. How to calculate the input image size for each layer?
10. How to build a GAN model from scratch in Pytorch?
11. How to train the model on Colab with GPU?
12. What are the challenges in GAN?
13. Applications of GAN
14. How to generate a fake image by using the Generator?